



COURSEWARE

Agentic AI Applications with Codex

Lesson Plan

Course Code	C695
Version	v1.0
Release Date	20 September 2026
Training Provider	Tertiary Infotech Academy Pte. Ltd. (UEN 201200696W)
Trainer	Dr Alfred Ang

Controlled document. Verify the version before use.

Document Control

Document	Version	Date	Owner	Change summary
Lesson Plan	1.0	20 September 2026	Tertiary Infotech Academy Pte. Ltd.	Course identity updated to C695; one-day schedule and learner resources aligned

Contents

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1. Course and Delivery Information

Course	Agentic AI Applications with Codex
Course code	C695
Duration	1 day / 7.5 instructional hours
Delivery mode	Facilitated classroom or synchronous learning with a local Codex learner sandbox
Required resources	215-slide deck, Learner Guide, ten lab folders, Git, Python 3, Codex CLI/app or IDE extension, browser for Lab 06

2. Learning Outcomes

- LO1 Analyse agentic AI applications, their strengths, limitations and suitability for a work problem.
- LO2 Correlate agent-control design with algorithm efficiency and evaluate the resulting process improvement.
- LO3 Assess methods for evaluating the effectiveness, safety and reliability of Codex-enabled workflows.
- LO4 Evaluate comparative effectiveness and recommend a governed Codex implementation using observable evidence.

3. Facilitation Principles

- Use the PPT for mechanism visualisation, technical dialogue and evidence interpretation; do not demonstrate click-by-click procedures from the slides.
- Use the Learner Guide and lab PDFs for detailed procedures, prompts, troubleshooting and acceptance checks.
- Keep all external, destructive and repository-publishing actions outside learner scope unless the trainer stages a safe simulation.
- Require observable execution evidence before accepting an agent response or learner conclusion.

4. One-Day Detailed Schedule

Time	Topic / activity	Method	Resources	Evidence / outcome
09:30–09:45	Course orientation, safety boundary and evidence standard	Facilitated briefing	Slides 1–9	Environment readiness and learner goal
09:45–10:45	Topic 1 + Labs 01–02: surfaces, agentic loops and prompt-to-plan delivery	Mechanism dialogue + guided practice	Slides 10–58; Lab 01–02 folders	Repository map, scoped prompt and passing checks
10:45–12:00	Topic 2 + Labs 03–05: instructions,	Technical demonstration +	Slides 59–110; Lab 03–05 folders	Instruction trace, permission log

Time	Topic / activity	Method	Resources	Evidence / outcome
	sandbox, approvals and worktrees	guided lab		and isolated commits
12:00–12:30	Lunch	—	—	—
12:30–14:00	Topic 3 + Labs 06–08: browser QA, MCP, skills, subagents and headless execution	Mechanism dialogue + guided lab	Slides 111–162; Lab 06–08 folders	Browser trace, reusable skill and JSONL contract
14:00–15:30	Topic 4 + Labs 09–10: golden tasks, comparative evaluation and governed adoption	Evidence analysis + guided capstone	Slides 163–211; Lab 09–10 folders	Evaluation report and bounded recommendation
15:30–17:00	Integrated capstone: inspect, plan, act, verify and correct	Independent challenge + peer explanation	All lab evidence; Slides 212–215	Working change, reviewable evidence pack and recovery path
17:00–17:30	Technical recap, transfer plan and course feedback	Facilitated reflection	Learner Guide; feedback slide	Personal transfer commitment and improvement feedback

5. Trainer Preparation Checklist

- Confirm the course identity, version and 215-slide deck before class.
- Verify that every lab folder contains README/PDF, AGENTS.md, prompt, starter, solution, verifier and evidence checklist/PDF.
- Use a learner-safe Codex environment; keep external, destructive and repository-publishing actions out of scope.
- Test repository, Codex and browser access without exposing credentials on projected screens.

6. References

- **Official course page:** <https://www.tertiarycourses.com.sg/agent-ai-applications-with-codex.html> — course identity, duration, learning outcomes and delivery context
- **OpenAI Codex learning hub:** <https://developers.openai.com/learn/codex> — current Codex workflows, guides and reference entry points
- **Codex cloud documentation:** <https://learn.chatgpt.com/docs/cloud> — cloud environments, parallel tasks and review workflow

- **OpenAI Academy: Codex:** <https://openai.com/academy/codex/> — official learning resources and demonstrations
- **OpenAI Codex product:** <https://openai.com/codex/> — Codex product surfaces and capabilities
- **OpenAI Codex repository:** <https://github.com/openai/codex> — CLI installation, repository guidance and source reference
- **AIFLOXIUM Codex guide:** <https://www.aifloxiium.online/blog/openai-codex-tutorial-complete-guide-2026> — supplementary workflow coverage
- **QCode Codex IDE guide:** <https://docs.qcode.cc/en/docs/ide/codex> — supplementary IDE integration perspective
- **DataCamp Codex tutorial:** <https://www.datacamp.com/tutorial/openai-codex> — supplementary cloud-task examples
- **YouMind Codex tutorial:** <https://youmind.com/landing/x-viral-articles/master-codex-ai-features-tutorial> — supplementary feature overview
- **AirMore Codex beginners review:** <https://airmore.ai/ai-review/codex-tutorial-beginners> — supplementary beginner perspective
- **Gauraw Codex tutorial:** <https://www.gauraw.com/openai-codex-tutorial/> — supplementary workflow and troubleshooting coverage
- **DeepWiki Codex interfaces:** <https://deepwiki.com/openai/codex/4-user-interfaces> — supplementary interface taxonomy